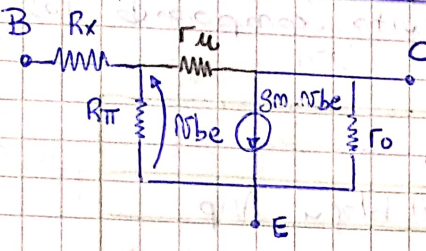
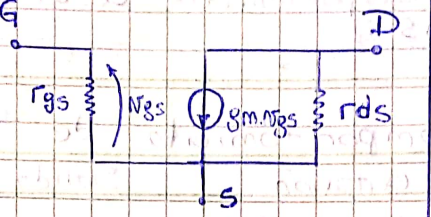
Resumen

TBJ



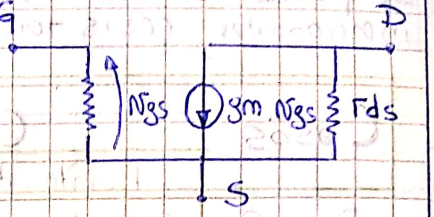
- $g_m = \frac{I_{CQ}}{V_{th}}$
- $r_\pi = \beta / g_m$
- $r_o = \frac{V_A}{I_{CQ}}$
- $r_\mu = \beta . r_o$

MOSFET



- $g_m = 2\sqrt{k . I_{DQ}} = 2k . V_{GS_{AR}}$
- $r_{ds} = 1 / \lambda . I_{DQ}$

JFET



- $g_m = 2 . \frac{I_{DSS}}{V_P} \left(1 - \frac{V_{GS}}{V_P}\right)$
- $r_{ds} = 1 / \lambda . I_{DQ}$

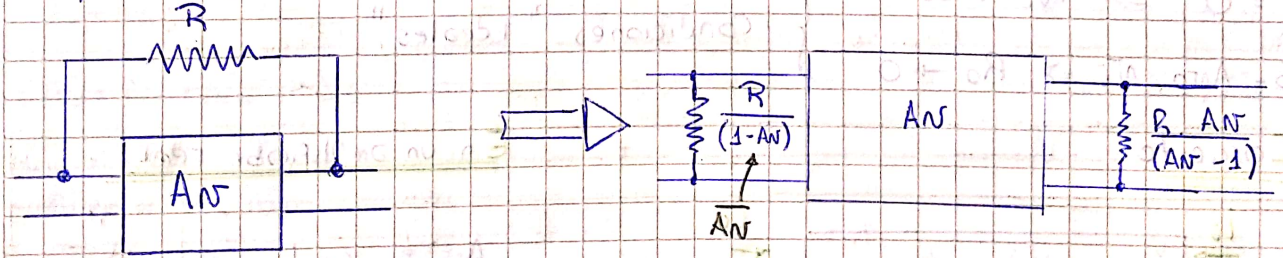
* $g_{m\,TBJ} >> g_{m\,MOS}, g_{m\,JFET}$ (SIEMPRE)

Frecuencias medias (CAPACITORES EQUIVALENCIAS)

Es el rango de Frecuencias para el cual los capacitores internos de los Transistores tienen reactancia mucho mayor a las resistencias del circuito, por lo que pueden considerarse circuitos abiertos (—o—); y los capacitores externos a los transistores tienen reactancias mucho menores que las resistencias del circuito, por lo que se consideran cortocircuitos.

Teorema de Miller

Dado un bloque Amplificador con ganancia A_V y una cierta resistencia R; si se conecta la resistencia entre los terminales no comunes del bloque amplificador, se puede realizar la siguiente equivalencia.



El teorema ampliado permite, en caso de tener un capacitor en lugar de una resistencia, obtener impedancias conectada en entrada y salida del amplificador Segun:

$$Z_{C_{IN}} = \frac{1/j\omega C}{(1 - A_V)} = \frac{1}{j\omega C(1 - A_V)} \qquad Z_{C_{OUT}} = \frac{1/j\omega C \cdot A_V}{(A_V - 1)} = \frac{A_V}{j\omega C(A_V - 1)}$$

Circuitos equivalentes

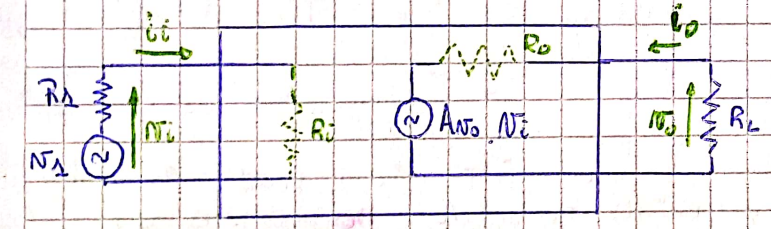
Dados R_s (resistencia serie del generador / excitador / fuente de excitación), R_i (resistencia de entrada del bloque amplificador), R_L (resistencia que carga la salida) y R_o (resistencia de salida del bloque amplificador); se puede analizar el comportamiento de un circuito comparando las mencionadas resistencias:

Casos	Comportamiento de la	
	Fuente de excitación	Salida del bloque Amp
$R_s \ll R_i$	generador de Tensión ideal	
$R_s \gg R_i$	generador de Corriente ideal	
$R_o \ll R_L$		generador de tensión controlado
$R_o \gg R_L$		generador de corriente controlado

Estos relaciones permiten dar una idea de qué amplificador ideal se trata.

Amplificadores Ideales

- **Tensión**



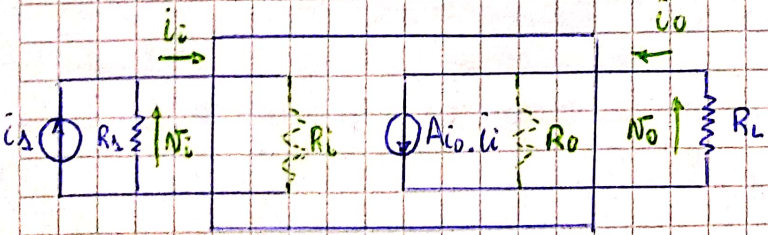
En un amplificador real:

$$A_{No} = \frac{N_o}{N_i}\Big|_{i_o=0}$$

- $i_i = 0 \Rightarrow R_i \to \infty$
- $N_o = A_{No}.N_i \Rightarrow R_o \to 0$

} Condiciones "ideales"

- **Corriente**



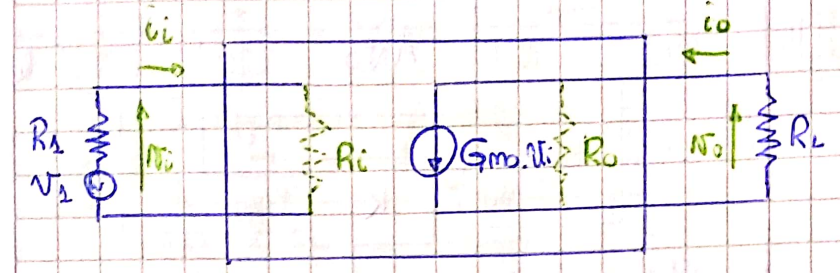
En un amplificador real:

$$A_{io} = \frac{i_o}{i_i}\Big|_{N_o=0}$$

- $N_i = 0 \Rightarrow R_i \to 0$
- $i_o = A_{io}.i_i \Rightarrow R_o \to \infty$

} Condiciones "ideales"

• Transconductancia

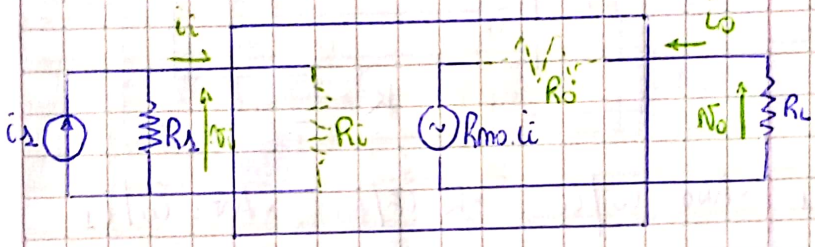


- $i_i = 0 \Rightarrow R_i \to \infty$
- $i_o = G_{mo}.V_i \Rightarrow R_o \to \infty$

En un amplificador real

$$G_{mo} = \left.\frac{i_o}{V_i}\right|_{V_o=0}$$

• Transresistencia

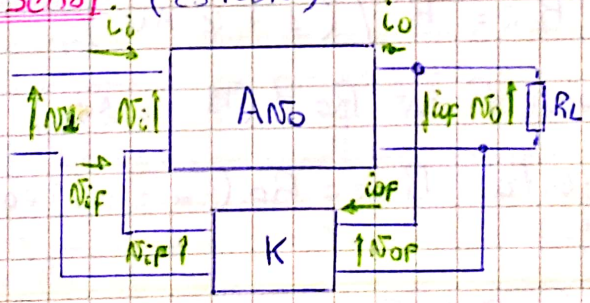


- $V_i = 0 \Rightarrow R_i \to 0$
- $i_o = R_{mo}.i_i \Rightarrow R_o \to 0$

En un amplificador real

$$R_{mo} = \left.\frac{V_o}{i_i}\right|_{i_o=0}$$

Realimentación de señal (estable)



- $A_{V_o} = V_o / V_i$
- $K = V_{if} / V_{OF}$
- $A_V = V_o / V_1$

• Negativa

- $|1 + A_{V_o}.K| > 1$
- $V_1 = V_i + V_{if} > V_i$, $A_V < A_{V_o}$
- $V_{if} = K.V_{OF}$
- $V_1 = V_i + V_{if} = V_o/A_{V_o} + KV_o \Rightarrow A_V = A_{V_o} / (1 + K A_{V_o})$

• Positiva

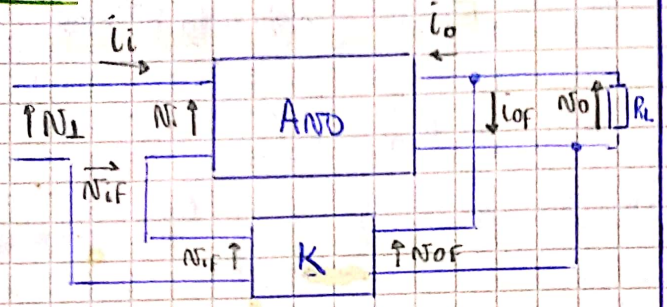
$|1 + A_{V_o}.K| < 1$

- Si $|1 + A_{V_o}.K| = 0 \Rightarrow$ Oscilador (desconectando excitación).
- Si $|A_{V_o}.K| < 1 \Rightarrow$ empeoran ciertos parámetros (respuesta en frecuencia, aumenta la distorsión).

Tipos de Realimentación

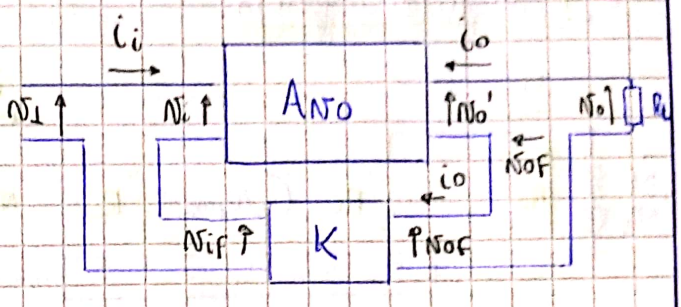
{ M: Muestreo
{ S: Suma

MVSV



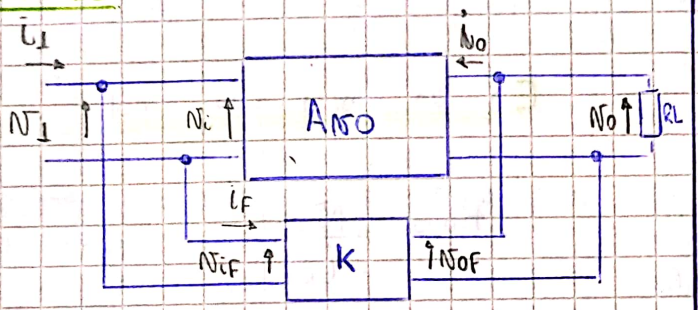
- $A_{VO} = V_o / V_i$
- $K = V_{iF} / V_{oF}$
- $A_V = V_o / V_i$

MISV



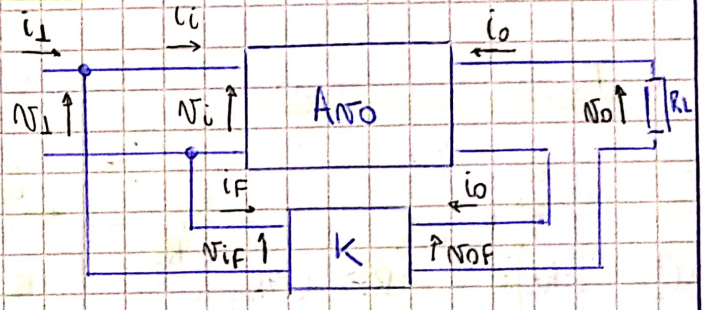
- $A_{VO} = i_o / V_i$
- $K = V_{iF} / i_o$
- $A_V = i_o / V_i$

MVSI



- $A_{VO} = V_o / i_i$
- $K = i_F / V_{oF}$
- $A_V = V_o / i_i$

MISI



- $A_{VO} = i_o / i_i$
- $K = i_F / i_o$
- $A_V = i_o / i_i$

Parametros de realimentación negativa

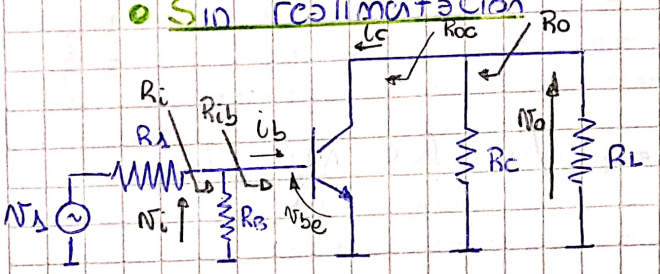
- Suma Tensión $R_{iR} = R_i \cdot (1 + K \cdot A_{VO})$
- Suma Corriente $R_{iR} = R_i / (1 + K \cdot A_{VO})$
- Muestreo de Tensión $R_{oR} = R_o / (1 + K \cdot A_{VO})$
- Muestreo de corriente $R_{oR} = R_o \cdot (1 + K \cdot A_{VO})$

	Variación con respecto a parm. sin realimentar		Tendencia del bloque Amp
	R_{iR}	R_{oR}	
MVSV	↑	↓	A_V
MVSI	↓	↓	R_m
MISV	↑	↑	G_m
MISI	↓	↑	A_i

Configuraciones de transistores

Emisor Común

• Sin realimentación



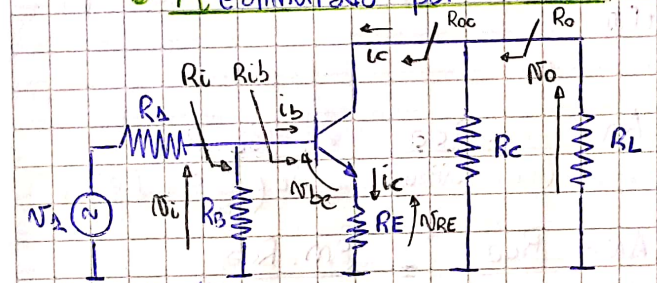
- $A_V = V_o / V_i = \dfrac{-i_c (R_c \parallel R_L)}{V_{be}} = -g_m \cdot R_{ca}$
- $R_i = R_B \parallel R_{ib}$

  $R_{ib} = r_\pi \Rightarrow R_i = R_B \parallel r_\pi$
- $R_o = R_{oc} \parallel R_c$

  $R_{oc} = r_o \rightarrow$ No se prende el generador de corriente. $\rightarrow$ $g_m \cdot V_{be}$ (VER)

  $R_o = R_c \parallel r_o \overset{\sim}{=} R_c$ (porque $r_o \gg R_c$)

• Realimentado por emisor

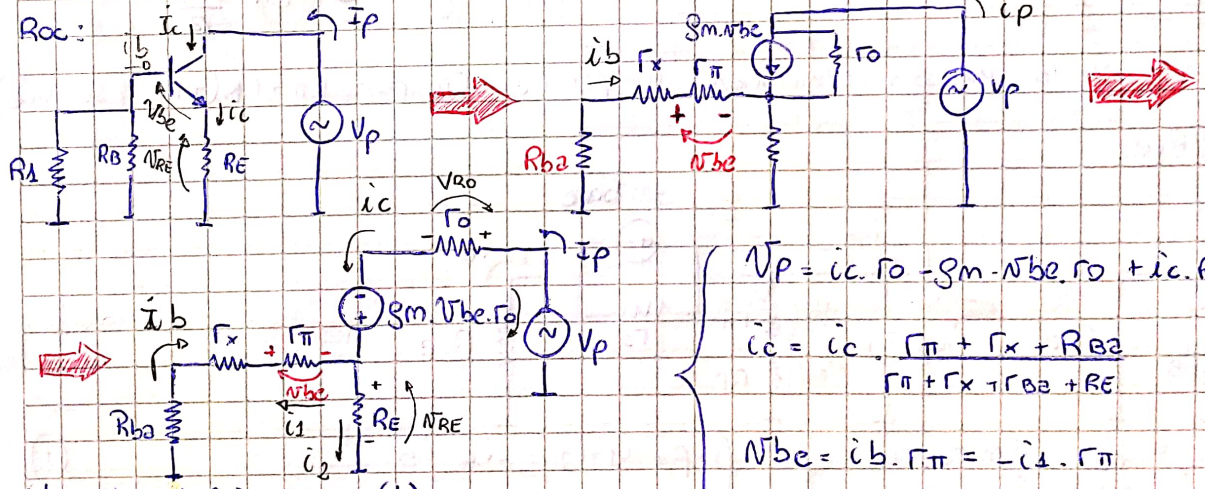


- $A_V = V_o / V_i = \dfrac{-i_c (R_c \parallel R_L)}{V_{be} + V_{RE}} = \dfrac{-i_c \cdot R_{ca}}{i_c \left(\frac{V_{be}}{i_c} + R_E \right)}$

  $A_V = -R_{ca} / \left(\frac{1}{g_m} + R_E \right)$
- $R_i = R_{ib} \parallel R_B$

  $R_{ib} = r_\pi + \beta \cdot R_E \Rightarrow R_i = R_B \parallel (r_\pi + \beta R_E)$ (VER)
- $R_o = R_c \parallel R_{oc}$

R_{oc}:



$$\begin{cases} V_p = i_c \cdot r_o - g_m \cdot V_{be} \cdot r_o + i_c \cdot R_E & (1) \\ i_c = i_c \cdot \dfrac{r_\pi + r_x + R_{Ba}}{r_\pi + r_x + r_{Ba} + R_E} & (2) \\ V_{be} = i_b \cdot r_\pi = -i_1 \cdot r_\pi & (3) \\ i_1 = i_c \cdot \dfrac{R_E}{r_\pi + r_x + R_{Ba} + R_E} & (4) \end{cases}$$

Usando (2), (3), (4) en (1)

$$V_p = i_c \cdot r_o + \underbrace{r_o \cdot g_m \cdot r_\pi \cdot i_c \dfrac{R_E}{(r_\pi + r_x + R_{Ba} + R_E)}}_{\beta} + i_c \cdot \dfrac{R_E (R_{Ba} + r_\pi + R_E)}{(r_\pi + r_x + R_{Ba} + R_E)}$$

$$R_{oc} = \frac{V_p}{i_p} = \frac{V_p}{i_c} = r_o \left(1 + \beta \cdot \frac{R_E}{(R_E + R_{Ba} + r_x + r_\pi)} \right) + \underbrace{\frac{R_E (R_{Ba} + r_\pi + r_x)}{(R_{Ba} + R_E + r_\pi + r_x)}}_{\text{constante?}}$$

$$R_{oc} \overset{\sim}{=} r_o \left(1 + \beta \cdot \frac{R_E}{(R_{Ba} + r_x + r_\pi + R_E)} \right)$$

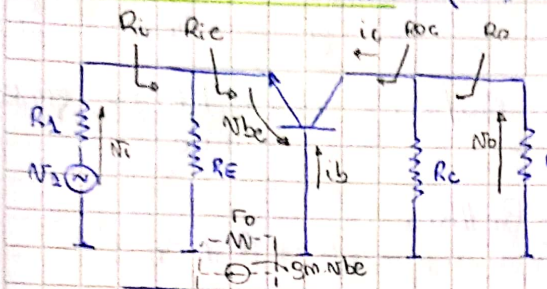
($r_o \gg$ Const)

$\rightarrow R_{oc} \overset{\sim}{=} r_o$, si $\beta \cdot R_E \ll 1$

$\rightarrow R_{oc} \overset{\sim}{=} r_o (1+\beta) \overset{\sim}{=} \beta \cdot r_o$, si $R_E \gg R_{Ba} + r_\pi + r_x$

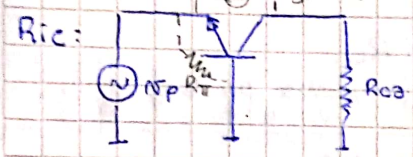
Base Común

- **Sin realimentación** ($r_x \approx 0\Omega$ Tomar $r_x \neq 0$ = realimentado x base)



- $A_V = V_o/V_i = \dfrac{-i_c (R_c \| R_L)}{-V_{be}} = g_m \cdot R_{ca}$ (gm = 1/rπ, Rca = Rc‖RL)
- $R_i = R_{ie} \| R_E$

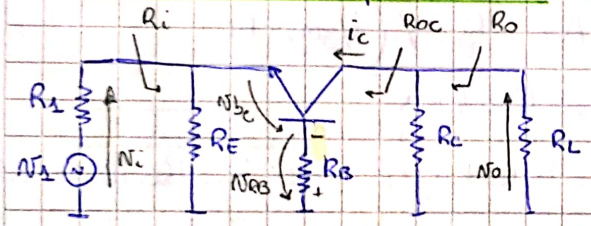
Ric:



$$\begin{cases} V_p = -V_{be} \\ i_p = -i_c \end{cases} \Rightarrow V_p/i_c = R_{ie} = \frac{V_{be}}{i_c} = 1/g_m$$

- $R_o = R_c \| R_{oc}$ R_{oc}: calculo similar a Emisor Común Realimentado
- $R_{oc} = r_o \left(1 + \dfrac{\beta \cdot R_1 \| R_E}{R_1 \| R_E + r_\pi} \right) + \dfrac{R_E \cdot r_\pi}{R_E + r_\pi}$

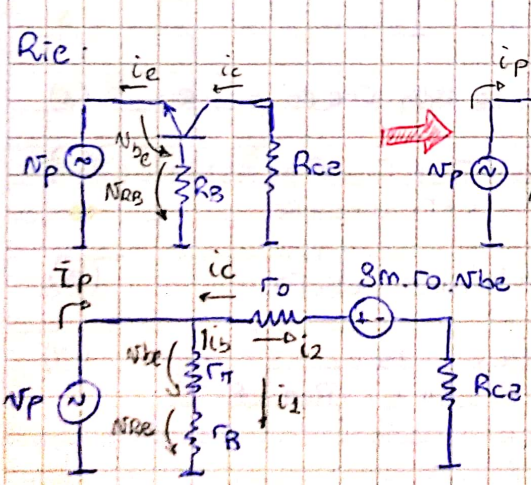
- **Realimentado por Base**



- $A_V = \dfrac{-i_c \cdot R_{ca}}{-(V_{be} + V_{RB})} = \dfrac{-i_c \cdot R_{ca}}{-i_c \left(\frac{V_{be}}{i_c} + \frac{R_B}{\beta} \right)}$
- $A_V = \dfrac{R_{ca}}{\frac{1}{g_m} + \frac{R_B}{\beta}} = \dfrac{g_m \cdot R_{ca}}{1 + g_m \frac{R_B}{\beta}}$

- R_o: mismo calculo que sin realimentacion, reemplazando r_π por $r_\pi + r_B (+ r_x$ si se pide)
- $R_i = R_E \| R_{ie}$

Rie:



$$\begin{cases} i_p = i_1 + i_2 = -i_c - i_b & (1) \\ i_1 = V_p / (r_\pi + R_B) & (2) \\ i_2 = (V_p - g_m \cdot R_{ca} \cdot V_{be}) / (r_o + r_{ca}) & (3) \\ V_{be} = -i_1 \cdot r_\pi & (4) \end{cases}$$

meto (2) en (4)

$$V_{be} = -r_\pi \cdot \frac{V_p}{r_\pi + R_B} \overset{(3)}{\Rightarrow} i_2 = \frac{V_p}{(r_o + r_{ca})} + \frac{(g_m r_o r_\pi) V_p}{(r_o + R_{ca}) \cdot (r_\pi + R_B)} \quad (5)$$

meto (3) y (2) en (1)

$$i_p = \frac{V_p}{r_\pi + R_B} + \frac{V_p}{r_o + R_{c2}} + \frac{\beta \cdot r_o \cdot V_p}{(r_o + R_{c2}) \cdot (r_\pi + R_B)}$$

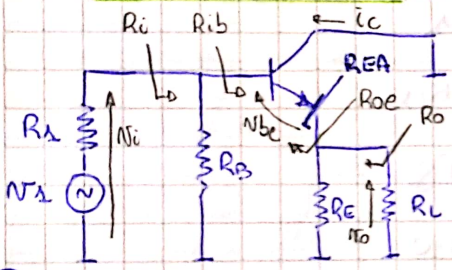
$$\frac{1}{R_{ie}} = \frac{1}{r_\pi + R_B} + \frac{1}{r_o + R_{c2}} + \frac{\beta \cdot r_o}{(r_o + R_{c2}) \cdot (r_\pi + R_B)}$$

$$(R_{ie})^{-1} = \frac{r_o + R_{c2} + r_\pi + R_B + \beta \cdot r_o}{(r_o + R_{c2})(R_B + r_\pi)} = \frac{r_o \cdot (r_\pi + R_B) + R_{c2}(r_\pi + R_B)}{r_o(1+\beta) + R_{c2} + r_\pi + R_B}$$

$$R_{ie} \underset{\uparrow r_o \to \infty}{\cong} \frac{r_\pi + R_B}{1+\beta} \underset{\uparrow \beta \gg 1}{\cong} \frac{r_\pi + R_B}{\beta}$$

Colector Común

- Sin realimentacion



$R_{E2} = R_E \parallel R_L$

$$A_V = \frac{i_c \cdot R_{E2}}{V_{be} + i_c \cdot R_{E2}} = \frac{i_c \cdot R_{E2}}{\left(R_{E2} + \frac{V_{be}}{i_c}\right) i_c} = \frac{R_{E2}}{R_{E2} + 1/g_m}$$

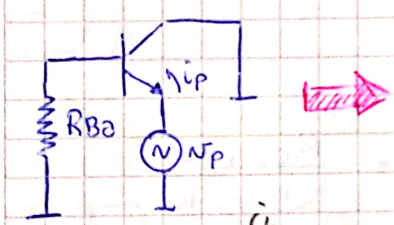
$$A_V = \frac{g_m \cdot R_{EA}}{1 + g_m \cdot R_{EA}}$$

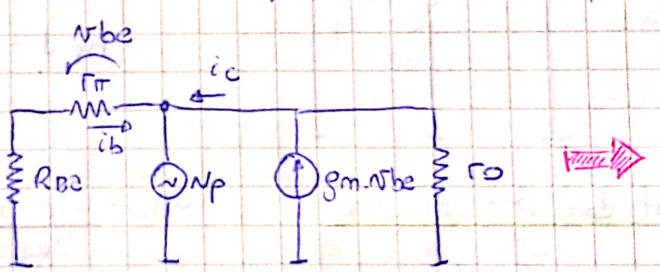
- $R_i = R_B \parallel R_{ib}$

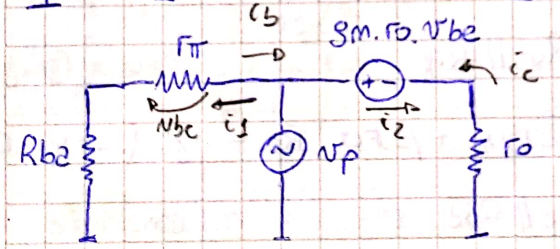
$R_{ib} = r_\pi + R_{E2} \cdot \beta$

- $R_o = R_E \parallel R_{oe} = R_E \parallel r_\pi/\beta$

Roe:







$$\begin{cases} \circ\ i_p = i_1 + i_2 & (1) \\ \circ\ i_1 = V_p / (r_\pi + R_{B2}) & (2) \\ \circ\ i_2 = (V_p - g_m \cdot r_o \cdot V_{be}) / (r_o) & (3) \\ \circ\ V_{be} = -i_1 \cdot r_\pi & (4) \end{cases}$$

meto (2) en (4)

$$V_{be} = \frac{-V_p \cdot r_\pi}{(r_\pi + R_{B2})} \overset{(3)}{\Rightarrow} i_2 = \frac{V_p}{r_o} + V_p \frac{(g_m \cdot r_\pi)}{r_\pi + R_{B2}} \quad (5)$$

meto (5) y (2) en (1)

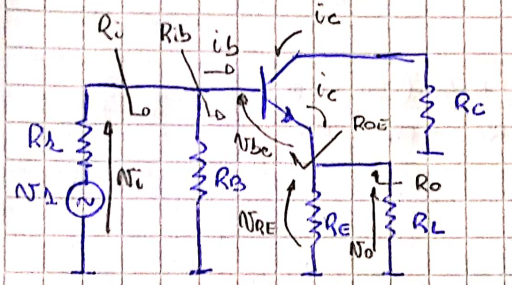
$$i_p = V_p \left(\frac{1}{r_\pi + R_{B2}} + \frac{1}{r_o} + \frac{\beta}{r_\pi + R_{B2}} \right) \Rightarrow \frac{1}{R_{oe}} = \frac{\beta + 1}{r_\pi + R_{B2}} + \frac{1}{r_o}$$

$$R_{oe} = \left(\frac{r_\pi + R_{B2}}{\beta+1} \,//\, r_o \right)$$

$$R_{oe} \overset{\uparrow}{\cong} \frac{r_\pi + R_{B2}}{\beta+1} \overset{\uparrow}{\cong} \frac{r_\pi + R_{B2}}{\beta}$$

$r_o \gg \frac{r_\pi + R_{B2}}{\beta+1}$ $\quad$ $\beta \gg 1$

- **Realimentado por Colector**



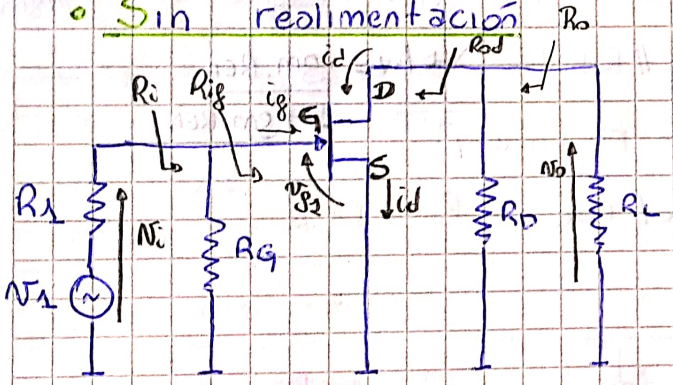
- $A_V = i_c \cdot R_{e2} / (V_{be} + i_c \cdot R_{e2}) = R_{E2} / \left(R_{e2} + 1/g_m\right)$
- $R_i = R_B \,//\, R_{ib}$

  $R_{ib} = r_\pi + R_{e2} \cdot \beta$
- $R_o \cong R_E \,//\, \left(\frac{r_\pi + R_{B2}}{\beta} \right)$

- Los calculos de R_i, R_o para los FET son analogos.

Source Comun

- **Sin realimentación**

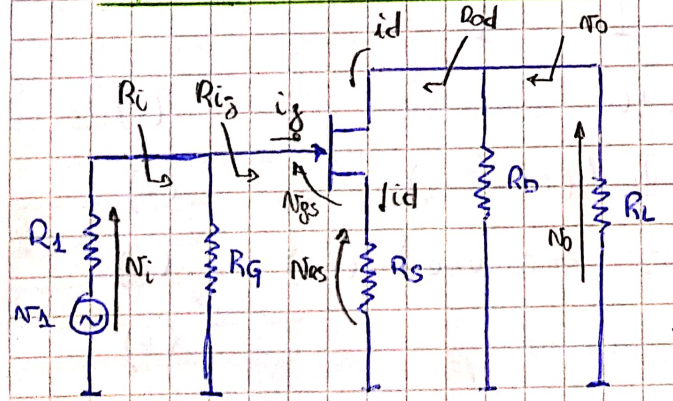


- $A_V = \frac{-i_d \cdot R_{D2}}{V_{g1}} = -g_m \cdot R_{D2}$
- $R_i = R_G \,//\, R_{ig}$

  $R_{ig} = r_{gs}$
- $R_o = R_o \,//\, R_{od}$

  $R_{od} = r_{ds}$

- **Realimentado por Source**



- $A_V = \frac{-i_d \cdot R_{D2}}{V_{g1} + V_{Rs}} = \frac{R_{D2}(-1)}{\frac{1}{g_m} + R_s}$
- $R_i = R_g \,//\, R_{ig}$

  $R_{ig} = r_{g1} + \beta_{FET} \cdot R_s = r_{gs}\left(1 + g_m \cdot R_s\right)$ $\quad$ $\beta_{FET} = r_{gs} \cdot g_m$
- $R_o = R_D \,//\, R_{od}$

$$R_{od} = r_{ds}\left(1 + \frac{g_m \cdot r_{gs} \cdot R_s}{R_{G2} + r_{gs} + R_s}\right) + R_s\left(\frac{r_{gs} + R_{G2}}{R_{G2} + r_{gs} + R_s}\right)$$

(constante)

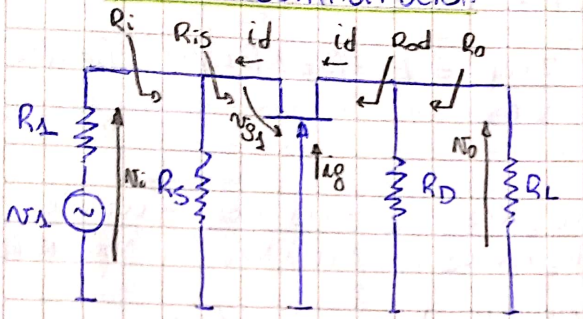
$R_{G2} = R_1 \,//\, R_G$

- $R_{od} \cong r_{ds}\left(1 + \frac{g_m \cdot r_{gs} \cdot R_s}{R_{G2} + r_{gs} + R_s}\right)$, $r_{ds} \gg$ const

  $\hookrightarrow R_{od} \cong r_{ds}\left(1 + g_m R_s\right)$, $r_{gs} \gg R_{G2} + \underbrace{r_{gs} + r_s}_{\text{Ver } g_3}$

Gate Común

Sin Realimentación



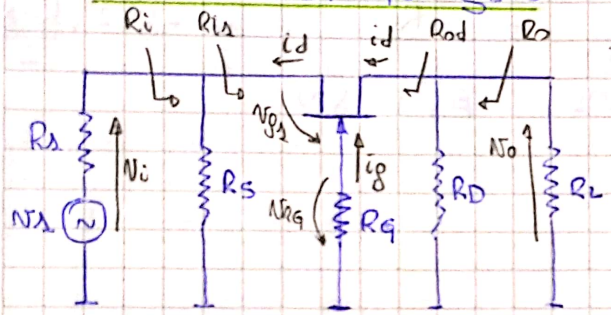
- $A_V = \frac{-i_d \cdot R_{D2}}{-V_{gs}} = g_m \cdot R_{D2}$
- $R_i = R_s \parallel R_{is}$

$R_{is} = 1/g_m$

- $R_{od} = r_{ds}\left(1 + \frac{g_m \cdot r_{gs}(R_{s2})}{r_{gs} + R_{s2}}\right) + r_{gs} \parallel R_s$ ($R_{s2} = R_s \parallel R_s$)
- $R_o = R_D \parallel R_{od}$

$R_{od} \cong (1 + g_m \cdot R_{s2})$

Realimentación por gate



- $A_V = \frac{-i_d \cdot R_{D2}}{-V_{gs} - V_{RG}} = \frac{-i_d \cdot R_{D2}}{\left(\frac{R_G}{\beta_{FET}} + \frac{V_{gs}}{i_d}\right)(-i_d)}$

$A_V = \frac{R_{D2}}{\frac{1}{g_m} + \frac{R_G}{\beta_{FET}}} \cong g_m \cdot R_{D2}$ ($\beta_{FET} \to \infty$)

- $R_i = R_s \parallel R_{is}$

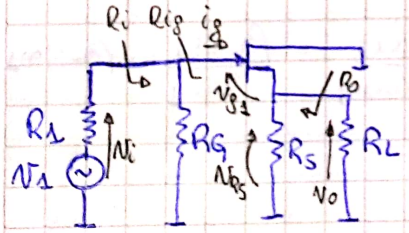
$R_{is} \cong \frac{r_{gs} + R_G}{1 + \beta_{FET}} \cong 1/g_m$

- $R_o = R_D \parallel R_{od}$

$R_{od} \cong r_{ds}(1 + g_m \cdot R_{s2})$

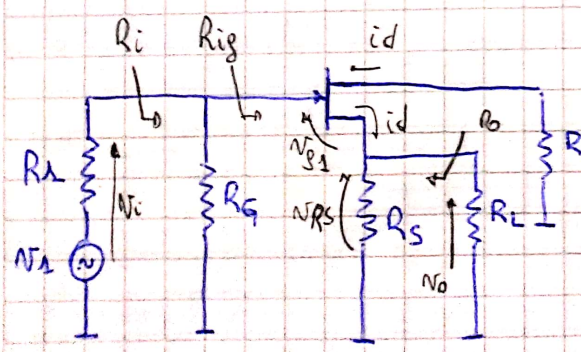
Drain Común

Sin realimentación



- $A_V = \frac{i_d \cdot R_{s2}}{i_d \cdot R_{s2} + V_{gs}} = R_{s2} \Big/ \left(1/g_m + R_{s2}\right)$
- $R_i = R_g \parallel R_{ig}$; $R_{ig} = r_{gs}(1 + g_m \cdot R_{s2})$
- $R_o = R_s \parallel R_{os}$; $R_{os} \cong \frac{r_{gs} + R_{GA}}{\beta_{FET}} \cong 1/g_m$

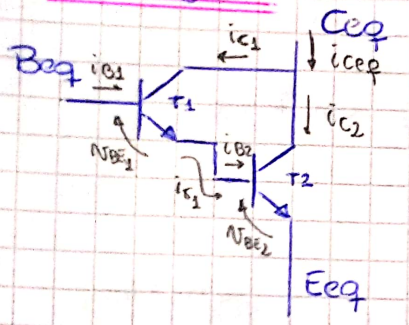
Realimentado por Drain



- $A_V = \frac{i_d \cdot R_{s2}}{i_d \cdot R_{s2} + V_{gs}} \cong R_{s2} \Big/ \left(R_{s2} + 1/g_m\right)$
- $R_i = R_G \parallel R_{ig}$ $\quad R_{ig} = r_{gs}(1 + g_m \cdot R_{s2})$
- $R_o = R_s \parallel R_{os}$

$R_{os} \cong 1/g_m$

Darlington



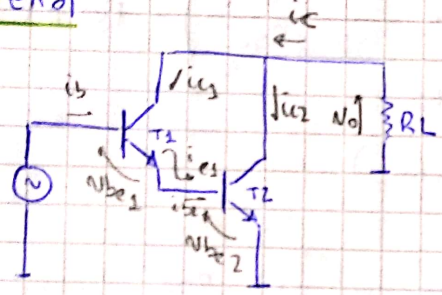
- $I_{ceq} = I_{C1} + I_{C2} = \beta_1 \cdot I_{B1} + \beta_2 \cdot I_{B2}$

$I_{ceq} = \beta_1 I_{B1} + \beta_2 (I_{B1} + I_{C1})$

$I_{ceq} = I_{B1} [\beta_1 + \beta_2 (1 + \beta_1)]$

$\beta^* = \frac{I_{ceq}}{I_{B1}} = \frac{I_{ceq}}{I_{Beq}} = \beta_1 + \beta_2 + \beta_1 \beta_2 \cong \beta_2 \cdot \beta_2$

Señal



- $A_V = \frac{V_o}{V_i} = \frac{-i_c \cdot R_{c2}}{V_{be1} + V_{be2}} = \frac{-i_{c1} \cdot R_{c2}}{V_{be1} + V_{be2}} = \frac{-i_{c2} \cdot R_{c2}}{V_{be1} + V_{be2}}$

$A_V = \frac{-i_{c2} \cdot R_{c2} (1 + 1/\beta_2)}{V_{be1} + V_{be2}} \cong \frac{-i_{c2} \cdot R_{c2}}{V_{be1} + V_{be2}}$

$A_V = \frac{-R_{c2}}{\frac{V_{be1}}{i_{c2}} + \frac{1}{g_{m2}}} = \frac{-R_{c2}}{\frac{1}{\beta \cdot g_{m1}} + \frac{1}{g_{m2}}} = -g_{m2} \cdot \frac{R_{c2}}{2}$

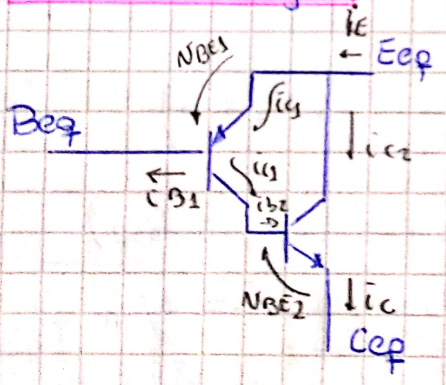
- $g_m^* = \frac{i_c}{V_{be}} = \frac{i_{c1}}{2V_{be1}} + \frac{i_{c2}}{2V_{be2}} = \frac{g_{m1}}{2} + \frac{g_{m2}}{2}$

$g_m^* = \frac{g_{m1}}{2} + \beta \cdot \frac{g_{m1}}{2} \cong \beta \cdot \frac{g_{m1}}{2} = \frac{g_{m2}}{2}$

- $r_\pi^* = \frac{V_{be}}{i_b} \rightarrow \frac{2V_{be1}}{i_b} = 2 r_{\pi_1}$

- $r_o^* = \frac{2}{3} r_{o_2}$

Cuasi - Darlington



- $\beta^* \approx \beta_1 \cdot \beta_2$

- $g_m^* = g_{m2} = \frac{i_c}{V_{be}} = \frac{i_{c2}}{V_{be1}} = \frac{\beta \cdot i_{c1}}{V_{be1}} = \beta \cdot g_{m1}$

- $r_\pi^* = r_{\pi_1}$

- $r_o^* = \frac{r_{o2}}{2}$